



North South University

Department of Mathematics and Physics

Course Outline

Course Name	Calculus and Analytical Geometry - II
Course Code	MAT 130
Section	2
Semester	Summer 2026
Credit Hours	3 Credit
Pre-requisite	MAT 120

Course Coordinator	Dr. Md Sumon Hossain, Department of Mathematics & Physics sumon.hossain@northsouth.edu
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Instructor's Name	Dr. Khandker Farid Uddin Ahmed
Email	khandker.ahmed@northsouth.edu
Office Room	SAC 1173
Office Time	11:20 AM - 12:50 PM and 02:40 PM - 04:10 PM
Class Time & Room	01:00 PM – 02:30 PM (SAC 207)

Text Book:

Howard Anton, Bivens, Stephen Davis, "Calculus, Early Transcendentals", 10th edition.

Course Short Description:	This course provides students an overview of the basic principle of the integral, the methodology of finding the area between curves, the length of a plane curve, surface area, and volume by revolving plane curves using integration.
Course Objectives:	1. To classify different types of proper and improper integrals and find the appropriate techniques for finding values of integrals, and to analyse the area of bounded and unbounded regions. 2. To analyse the basic geometric properties of conic sections parabolas, ellipses, and hyperbolas. 3. To demonstrate student' understanding of the relationship between the exponential and hyperbolic functions, their graphs and the application of hyperbolic functions in the real-life problems. 4. To develop the ability to apply the basic principles of integration to find the length of a curve, surface area of revolution, area between two curves and the volume of solids generated by revolution of curves.

Course Learning Outcomes	Upon successful completion of this course, students will be able to (CO-1) Classify the type of a given integral and apply the appropriate technique for finding the value of the integral. (CO-2) Formulate and evaluate integrals to find the length of curves, the area between curves, the area of unbounded regions, and the area of surfaces of revolution. (CO-3) Analyze the structure of solids generated by revolution of a region bounded by curves to evaluate volume. (CO-4) Illustrate parametric curves and conic sections, and analyze their various properties. (CO-5) Develop the ability to apply polar coordinates to find the area of regions bounded by polar curves.
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Marks Distribution: (Subject to change according to the directives from UGC/NSU)

Assessment Strategy and Grading Scheme	
Grading tool	Marks
Attendance and class performances	05%
Assignments (At least 2 assignments)	10%
Quizzes (Best 2 quizzes out of at least 3 quizzes)	20%
Midterm	30%
Final Exam	35%

Grading Policies: As per NSU grading policy.

Attendance Policy: As per NSU policy.

Academic Integrity Policy: The Department of Mathematics and Physics does not tolerate academic dishonesty by its students. At a minimum, students must not be involved in cheating, copyright infringement, submitting the same work in multiple courses, significant collaboration with other individuals outside of sanctioned group activities, or fabrication. Students are advised that violations of the Student Integrity Code will be treated seriously, with special attention given to repeated offenses.

Please see the NSU Student Handbook, Sections: “Disciplinary Actions” and “Procedures and Guidelines”.

Makeup and Exam policy: NO makeup for quizzes and NO Formative assessment will be retaken under any circumstances. If a student misses the Midterm and/or Final exams due to circumstances beyond their control (official valid documents are required) and is informed beforehand (if possible), reasonable arrangements may be considered. **Students may get to see/recheck their midterm and Final exam scripts.**

Mapping of Course Outcomes

CLOs	Course Outcomes (CO)	Bloom’s taxonomy domain/level (C: Cognitive P: Psychomotor A: Affective)	Delivery methods and activities	Assessment tools
CO-1	Classify the type of a given integral and apply the appropriate technique for finding the value of the integral.	C2, C3	Lecture Group work	Quiz Midterm exam

CO-2	Formulate and evaluate integrals to find the length of curves, the area between curves, the area of unbounded regions, and the area of surfaces of revolution.	C3, C4, P2	Lecture Group work	Midterm exam Assignment
CO-3	Analyze the structure of solids generated by the revolution of a region bounded by curves to evaluate volume.	C3, C4, P2	Lecture Discussion	Group work Quiz Assignment
CO-4	Illustrate parametric curves and conic sections, and analyze their various properties.	C4, P2	Lecture Discussion	Assignment Final Exam

Contents

Lecture	Topics	Article no in the textbook	Assessment tools	Learning Outcomes
1, 2, 3	Integration by Substitution Integration by parts, Trigonometric integrals, Wallis' Sine, Cosine formulas	7.1, 7.2, 7.3	Quiz1, Discussions Midterm	CO-1
4, 5, 6	Trigonometric substitutions, Integrating Rational functions by partial fractions, Hyperbolic functions, and hanging cables,	7.4, 7.5	Quiz 1, Assignment I Midterm	CO-1
7, 8, 9,	Area between two curves, Length of a plane curve	6.1, 6.4	Quiz 2 Discussions Midterm	CO-2 CO-3
10, 11	Area of a surface of revolution, Hyperbolic functions, and hanging cables	6.5, 6.9	Quiz 2 Discussions Midterm	CO-2 CO-3
12	Midterm Examination			
13, 14, 15	Volumes by slicing disks, Volumes by slicing washers, Volumes by Cylindrical shells	6.2, 6.3	Assignment II Final	CO-2, CO-3
16, 17, 18	Improper Integrals, Parametric equations, Tangent lines, and arc length for parametric curves	7.8, 10.1	Quiz 3 Final	CO-3
19, 20, 21	Polar coordinates, Tangent lines, Arc length, and area of polar curves	10.2, 10.3	Quiz 3 Final	CO-4
22, 23, 24	Conic sections: parabola, ellipse, Hyperbola, Conic sections in polar coordinates	10.4, 10.6	Final Assignment II	CO-5
Final Exam (Declared by the Controller of Examinations)				